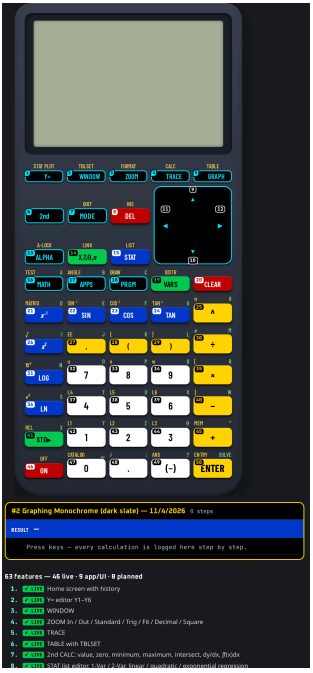
U-45 CALCULATORS

User Manual

Calculator #2 Graphing Monochrome (dark slate)

Graphing – monochrome



Complete kit: functions · numbered key map · worked examples · tricks & hacks · shortcuts · common mistakes · practice drills

About this manual

This manual belongs to the **U-45 CALCULATORS** app and describes calculator **#2 — Graphing Monochrome (dark slate)** exactly as it behaves in the app. Every worked example and every drill answer in this book was run on the app's own keys by an automated checker before printing; only results that matched were kept.

How key presses are written

Keys are written as a chip showing the **key number** and its face, for example  . Numbers follow the numbering printed on the keys (switch it on with "# On"). A shifted function is written as two chips: the shift key, then the key that carries the function.

Entry style of this model

algebraic with history (type the expression, then EXE / ENTER).

Chapters

1. Getting started
2. Features
3. Numbered key map
4. Key-by-key reference
5. Worked examples
6. Tricks & hacks
7. Keyboard shortcuts
8. Common mistakes
9. Practice drills
10. Appendix A — Drill answers
11. Appendix B — Function finder (A–Z)
12. Appendix C — Glossary

13. Appendix D — Model notes & limits

14. Appendix E — Quick-reference card

1 · Getting started

Open U-45 CALCULATORS, choose **#2** from the home screen (or the jump list at the top), and the calculator appears with its numbered, colour-coded keys. Click or tap keys, or type on a keyboard.

Numbers on every key

Every key carries a small number in the same reading order as this manual (left → right, top → bottom; the navigation pad counts ▲ ▼ ◀ ▶ then its centre). Use "# On / # Off" in the bar above the calculator to show or hide them — the setting is remembered.

High-contrast colours

Keys are coloured by job: white = digits, yellow = operators/execute, blue = functions, green = memory/variables, red = clear/delete/power, black with cyan = shift/modes/menus/navigation. Press "Classic" to return to the original model colours and "Hi-Contrast" to come back.

Three typefaces

Display text uses the U45 Display face, key legends use U45 Keys (a condensed face that stays readable on small keys) and key numbers / helper text use U45 Mono. All three are bundled inside the app, so they work offline.

Copy the display

The Copy button copies exactly what the display shows (indicators that are switched off are skipped) so you can paste a result into a message or spreadsheet.

Fit or 1:1

Fit scales the calculator to your screen; 1:1 shows it at its design size. On a phone, Fit is the comfortable choice.

Reset

Reset rebuilds this calculator from scratch: display, memory, stack and modes return to their start-up state. Use it before each worked example in this manual.

Keyboard

On a computer you can type: digits, + − * / ^ ( ), Enter or = to execute, Backspace to delete, Esc to clear, arrows to move. Letters reach common functions (see the Keyboard shortcuts chapter).

Tooltips

Hover over (or long-press) a key to see "#number name".

Tools rail (v1.4)

A cyan icon rail on the right of every calculator opens 15 panels: Currency (166 currencies, rates auto-updated from open.er-api.com or Frankfurter when online and cached with a date; the bundled dated snapshot is used offline), Units (exact SI factors), Health (published formulas, not medical advice), Simple, Complex, Loans, Construction, Mechanical and Engineering (30+ numbered tools each), Calendar generator, Customize (26 options, saved per calculator), Settings (31 options), History, Hide and View. On narrow phones the rail shows icons only.

History (v1.4)

Every calculation is logged per calculator with a date and time. Open History on the rail to search, re-use (types the input back into the calculator), copy, delete one, select several and delete, or delete all (asks for confirmation). History is stored encrypted on this device (AES-GCM); exporting it needs the passcode.

Camera Solve (v1.4)

The green CS button between Restore and Social reads a problem from the camera, a photo, a PDF or a text file. Recognition (Tesseract OCR, English) runs offline on the device; an online language download is only used if you switch it on in Settings. Check and edit the recognised text, then Solve shows the steps offline. OCR is not available when the app is opened as a plain file:// page in a browser.

Layouts and view modes (v1.4)

The layout selector between Fit and # offers Auto plus 12 layouts: 3D classic, 3D high-contrast, Classic, Pocket, Compact, Expanded, Desktop wide, Split, One-hand left, One-hand right, Large print and Minimal. Auto picks one for your screen and your choice is remembered per calculator. The View panel hides the top bars, the tool bar, the rail or everything (full screen); the round yellow ⤢ button at bottom-left (or Esc) always restores the normal view.

Hide items (v1.4)

The Hide panel lists every button, rail icon and section of this calculator as tick boxes. Hiding is instant; unhiding asks for the app passcode.

Image protection (v1.4)

Right-click, drag and long-press saving of images is blocked inside the app. Saving or sharing an image export (PNG, JPG, QR, calendar), printing a calendar, and downloading or sharing the app itself ask for the passcode, and exported images carry a U-45 watermark. Honest note: this cannot stop a screenshot or screen recording made by the operating system or another device.

Works offline

U-45 CALCULATORS is an offline app: after the first load everything — including fonts — comes from the device.

Colour groups on this calculator

Colour	Group	Keys on this model
num	Digits & number entry	#32, #33, #34, #37, #38, #39, #42, #43, #44, #47, #48, #49
op	Operators & execute	#25, #27, #28, #29, #30, #35, #40, #45, #50
fn	Functions	#15, #21, #22, #23, #24, #26, #31, #36
mem	Memory, variables & registers	#14, #19, #41
clr	Clear, delete & power	#8, #20, #46
mod	Shift, modes, menus & navigation	#1, #2, #3, #4, #5, #6, #7, #9, #10, #11, #12, #13, #16, #17, #18

2 · Features

63 features: 46 live on the keys, 9 app/UI features, 8 planned (documented but not yet working — they are clearly marked and never used in examples).

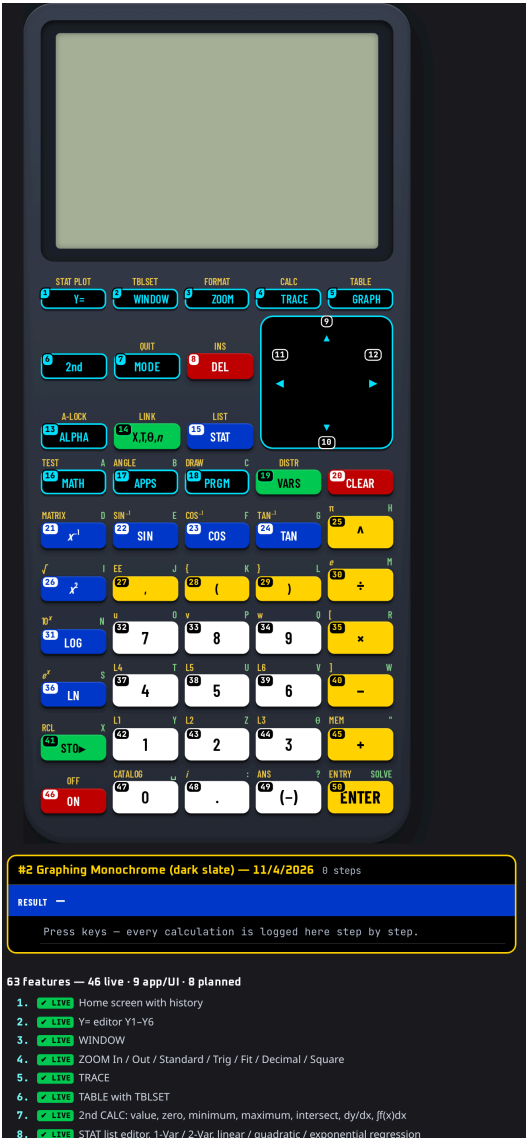
1. LIVE **Home screen with history**
2. LIVE **Y= editor Y1–Y6**
3. LIVE **WINDOW**
4. LIVE **ZOOM In / Out / Standard / Trig / Fit / Decimal / Square**
5. LIVE **TRACE**
6. LIVE **TABLE with TBLSET**
7. LIVE **2nd CALC: value, zero, minimum, maximum, intersect, dy/dx, ∫f(x)dx**
8. LIVE **STAT list editor, 1-Var / 2-Var, linear / quadratic / exponential regression**
9. LIVE **STAT PLOT scatter**
10. LIVE **MATH menu: fractions, $\sqrt[3]{}$, abs, round, nDeriv, fnInt**
11. LIVE **MATRIX**
12. LIVE **DISTR normal / binomial / Poisson**
13. LIVE **ALPHA SOLVE numeric solver**
14. LIVE **Variables A–Z and θ**
15. LIVE **Square root √**
16. LIVE **Square x^2**
17. LIVE **Reciprocal x^{-1} (1/x)**
18. LIVE **Power x^y / ^ with right-to-left chaining (2^3^2 = 512)**
19. LIVE **Common logarithm $\log_{10}$**
20. LIVE **Natural logarithm ln**

21. LIVE **Antilog 10^x**
22. LIVE **Exponential e^x**
23. LIVE **Trigonometry sin / cos / tan honouring the current angle unit**
24. LIVE **Inverse trigonometry $\sin^{-1}$ / $\cos^{-1}$ / $\tan^{-1}$ with domain checking (Math ERROR outside −1…1)**
25. LIVE **Angle units DEG / RAD / GRA with unit-suffix conversion (°, ʳ, ᵍ)**
26. LIVE **π constant to full internal precision**
27. LIVE **Euler e constant**
28. LIVE **Scientific-notation entry ×10^x / EXP / EEX**
29. LIVE **Sign change (−) / +/− / CHS**
30. LIVE **Ans recall of the last result; operators auto-insert Ans**
31. LIVE **Store a value into memory / a variable (STO, sto►, →, Min)**
32. LIVE **Recall memory / variables (RCL, MR, recall)**
33. LIVE **Named variables (A–F, X, Y, M … or A–Z) usable inside expressions**
34. LIVE **SOLVE — numeric equation solver (Newton / bisection)**
35. LIVE **TABLE — tabulate f(x) over start / end / step**
36. LIVE **Statistics: data entry, n, $\bar{x}$, σ, s, Σx, Σx^2**
37. LIVE **Two-variable statistics and linear regression (a, b, r)**
38. LIVE **Matrix mode: define, add, multiply, det, transpose, identity**
39. LIVE **Complex numbers: i, arg, conjugate, Re / Im, polar ↔ rectangular**
40. LIVE **MODE / SETUP: Fix / Sci / Norm display, angle unit, fraction format**
41. LIVE **Backspace / DEL editing of the current entry**
42. LIVE **INS insert mode for editing in the middle of an expression**
43. LIVE **Replay ▲ ▼ — scroll back to previous calculations and re-edit them**

44. LIVE CATALOG — alphabetical list of every function
45. LIVE MATH / Toolbox menu of templates and functions
46. LIVE Probability distributions (normal, binomial, Poisson) and PRB menu
47. APP/UI Every key numbered 1…N in reading order — matches the key map in the user manual
48. APP/UI High-contrast 6-colour key groups (digits, operators, functions, memory, clear, modes) with a “Classic” toggle
49. APP/UI Three bundled offline fonts: Oxanium (display/titles), Barlow Condensed (key faces), JetBrains Mono (key numbers & helpers)
50. APP/UI Show / hide key numbers with one button (remembered between sessions)
51. APP/UI Copy the current display text to the clipboard
52. APP/UI Physical-keyboard shortcuts: digits, + – * / ^ ( ) . Enter Backspace Esc arrows, s c t l n r ! % e p q i
53. APP/UI Fit / 1:1 zoom that never pushes keys off a phone screen
54. APP/UI Reference photo viewer and tool-tip name on every key
55. APP/UI Reset = fresh power-on state; works fully offline (PWA cache / Android APK)
56. PLANNED Programs (PRGM)
57. PLANNED LINK transfer
58. PLANNED List literals {…}
59. PLANNED Graph image export (PNG)
60. PLANNED Shareable function list link
61. PLANNED Dark screen theme
62. PLANNED Speech read-out of values

63. PLANNED Exam mode

3 · Numbered key map



Calculator #2 with key numbers shown (Hi-Contrast colours).

Key table

“Shift” = 2nd / f / SHIFT layer; “Alpha” = ALPHA layer; “3rd” = 3rd / g layer. A dot means the layer is empty on that key.

#	Face	Shift	Alpha	3rd	Group
1	Y=	STAT PLOT	·	·	mod
2	WINDOW	TBLSET	·	·	mod
3	ZOOM	FORMAT	·	·	mod
4	TRACE	CALC	·	·	mod
5	GRAPH	TABLE	·	·	mod
6	2nd	·	·	·	mod
7	MODE	QUIT	·	·	mod
8	DEL	INS	·	·	clr
9	▲	·	·	·	mod
10	▼	·	·	·	mod
11	◀	·	·	·	mod
12	▶	·	·	·	mod
13	ALPHA	A-LOCK	·	·	mod
14	X,T,θ,n	LINK	·	·	mem
15	STAT	LIST	·	·	fn
16	MATH	TEST	A	·	mod
17	APPS	ANGLE	B	·	mod
18	PRGM	DRAW	C	·	mod
19	VARS	DISTR	·	·	mem
20	CLEAR	·	·	·	clr
21	x^{-1}	MATRIX	D	·	fn
22	SIN	SIN^{-1}	E	·	fn
23	COS	COS^{-1}	F	·	fn

#	Face	Shift	Alpha	3rd	Group
24	TAN	TAN^{-1}	G	·	fn
25	^	π	H	·	op
26	x^2	√	I	·	fn
27	,	EE	J	·	op
28	(	{	K	·	op
29	)	}	L	·	op
30	÷	e	M	·	op
31	LOG	10x	N	·	fn
32	7	u	O	·	num
33	8	v	P	·	num
34	9	w	Q	·	num
35	×	[	R	·	op
36	LN	ex	S	·	fn
37	4	L4	T	·	num
38	5	L5	U	·	num
39	6	L6	V	·	num
40	−	]	W	·	op
41	STO▶	RCL	X	·	mem
42	1	L1	Y	·	num
43	2	L2	Z	·	num
44	3	L3	θ	·	num
45	+	MEM	"	·	op
46	ON	OFF	.	·	clr
47	0	CATALOG	␣	·	num
48	.	i	:	·	num
49	(−)	ANS	?	·	num
50	ENTER	ENTRY	SOLVE	·	op

4 · Key-by-key reference

Every key on calculator #2, in number order, with every function it carries in this app.

#1 Y= `Shift, modes, menus & navigation`

Primary: Y= — enter functions to graph

shifted (2nd / f / SHIFT layer) (press #6 first): STAT PLOT — STAT PLOT

#2 WINDOW `Shift, modes, menus & navigation`

Primary: WINDOW — set the graph view

shifted (2nd / f / SHIFT layer) (press #6 first): TBLSET — TBLSET — table start and step

#3 ZOOM `Shift, modes, menus & navigation`

Primary: ZOOM — zoom presets

shifted (2nd / f / SHIFT layer) (press #6 first): FORMAT — DRAW menu

#4 TRACE `Shift, modes, menus & navigation`

Primary: TRACE — move along a graph

shifted (2nd / f / SHIFT layer) (press #6 first): CALC — CALC — graph analysis (zero, min, max, intersect)

#5 GRAPH `Shift, modes, menus & navigation`

Primary: GRAPH — draw the graph

shifted (2nd / f / SHIFT layer) (press #6 first): TABLE — TABLE — table of values

#6 2nd `Shift, modes, menus & navigation`

Primary: SHIFT / 2nd — the next key uses its upper (shift-coloured) function

#7 MODE `Shift, modes, menus & navigation`

Primary: MODE — choose calculation mode

shifted (2nd / f / SHIFT layer) (press #6 first): QUIT — QUIT — leave the current screen / app

#8 DEL `Clear, delete & power`

Primary: DEL — delete the character at the cursor

shifted (2nd / f / SHIFT layer) (press #6 first): INS — INS — insert mode at the cursor

Keyboard: `Backspace` `Delete`

#9 ▲ `Shift, modes, menus & navigation`

Primary: Cursor ▲ — up / previous history line / previous item

#10 ▼ `Shift, modes, menus & navigation`

Primary: Cursor ▼ — down / next history line / next item

#11 ◀ Shift, modes, menus & navigation

Primary: Cursor ◀ — move left in the line

#12 ▶ Shift, modes, menus & navigation

Primary: Cursor ▶ — move right in the line

#13 ALPHA Shift, modes, menus & navigation

Primary: ALPHA — the next key types its letter/variable (red/green label)

shifted (2nd / f / SHIFT layer) (press #6 first): A-LOCK — ALPHA — the next key types its letter/variable (red/green label)

#14 X,T,θ,n Memory, variables & registers

Primary: The graphing variable (X / x,θ,t,n) for Y= functions, tables and solving

shifted (2nd / f / SHIFT layer) (press #6 first): LINK — Shows the message “Link not available” (feature not provided in this app)

#15 STAT Functions

Primary: STAT

shifted (2nd / f / SHIFT layer) (press #6 first): LIST — LIST menu

#16 MATH `Shift, modes, menus & navigation`

Primary: MATH menu — extra math functions

shifted (2nd / f / SHIFT layer) (press #6 first): TEST — Shows the message "TEST: type = < > with ALPHA" (feature not provided in this app)

alpha layer (press #13 first): A — Variable A (store with STO/ →, recall by name)

#17 APPS `Shift, modes, menus & navigation`

Primary: APPS — applications

shifted (2nd / f / SHIFT layer) (press #6 first): ANGLE — Angle symbol ∠ (polar complex form)

alpha layer (press #13 first): B — Variable B (store with STO/ →, recall by name)

#18 PRGM `Shift, modes, menus & navigation`

Primary: PRGM — programs

shifted (2nd / f / SHIFT layer) (press #6 first): DRAW — DRAW menu

alpha layer (press #13 first): C — Variable C (store with STO/ →, recall by name)

#19 VARS `Memory, variables & registers`

Primary: VARS — variables menu

shifted (2nd / f / SHIFT layer) (press #6 first): DISTR — DISTR — probability distributions

#20 CLEAR Clear, delete & power

Primary: CLEAR — clear the entry line / screen

Keyboard: Escape

#21 x^{-1} Functions

Primary: Reciprocal x^{-1} (1/x)

shifted (2nd / f / SHIFT layer) (press #6 first): MATRIX — MATRIX menu

alpha layer (press #13 first): D — Variable D (store with STO/ →, recall by name)

Keyboard: i

#22 SIN Functions

Primary: Sine (uses the current angle unit)

shifted (2nd / f / SHIFT layer) (press #6 first): SIN^{-1} — Inverse sine sin^{-1}

alpha layer (press #13 first): E — Variable E (store with STO/ →, recall by name)

Keyboard: s

#23 COS Functions

Primary: Cosine (current angle unit)

shifted (2nd / f / SHIFT layer) (press #6 first): COS^{-1} — Inverse cosine cos^{-1}

alpha layer (press #13 first): F — Variable F (store with STO/ →, recall by name)

Keyboard: c

#24 TAN Functions

Primary: Tangent (current angle unit)

shifted (2nd / f / SHIFT layer) (press #6 first): TAN^{-1} — Inverse tangent $\tan^{-1}$

alpha layer (press #13 first): G — Variable G (store with STO/ →, recall by name)

Keyboard: t

#25 ^ Operators & execute

Primary: Power x^y / ^ — raise to any power. Chains right-to-left (2^3^2 = 2^9 = 512)

shifted (2nd / f / SHIFT layer) (press #6 first): π — π (3.14159265…) as an exact constant

alpha layer (press #13 first): H — Variable H (store with STO/ →, recall by name)

Keyboard: ^

#26 x^2 Functions

Primary: Square x^2

shifted (2nd / f / SHIFT layer) (press #6 first): √ — Square root √

alpha layer (press #13 first): I — Variable I (store with STO/ →, recall by name)

Keyboard: q

#27 , Operators & execute

Primary: Comma / argument separator (between function arguments, list items or coordinates)

shifted (2nd / f / SHIFT layer) (press #6 first): EE — $\times 10^x$ / EXP / EE — enter a power-of-ten exponent (scientific notation), e.g. 6.02 $\times 10^x$ 23

alpha layer (press #13 first): J — Variable J (store with STO/ →, recall by name)

Keyboard: [,]

#28 (Operators & execute

Primary: Opening bracket (

shifted (2nd / f / SHIFT layer) (press #6 first): { — Types "{"

alpha layer (press #13 first): K — Variable K (store with STO/ →, recall by name)

Keyboard: [(]

#29) Operators & execute

Primary: Closing bracket). Closing brackets at the end of a line may be omitted

shifted (2nd / f / SHIFT layer) (press #6 first): } — Types "}"

alpha layer (press #13 first): L — Variable L (store with STO/ →, recall by name)

Keyboard: [)]

#30 ÷ Operators & execute

Primary: Division ÷

shifted (2nd / f / SHIFT layer) (press #6 first): e — Euler's number e (2.71828...)

alpha layer (press #13 first): M — Variable M (store with STO/ →, recall by name)

Keyboard: /

#31 LOG Functions

Primary: Common logarithm $\log_{10}$

shifted (2nd / f / SHIFT layer) (press #6 first): 10x — 10^x — antilog

alpha layer (press #13 first): N — Variable N (store with STO/ →, recall by name)

Keyboard: l

#32 7 Digits & number entry

Primary: Digit 7

alpha layer (press #13 first): O — Variable O (store with STO/ →, recall by name)

Keyboard: 7

#33 8 Digits & number entry

Primary: Digit 8

alpha layer (press #13 first): P — Variable P (store with STO/ →, recall by name)

Keyboard: 8

#34 9 Digits & number entry

Primary: Digit 9

alpha layer (press #13 first): Q — Variable Q (store with STO/ →, recall by name)

Keyboard: 9

#35 × Operators & execute

Primary: Multiplication ×

shifted (2nd / f / SHIFT layer) (press #6 first): [— Types "["

alpha layer (press #13 first): R — Variable R (store with STO/ →, recall by name)

Keyboard: * x

#36 LN Functions

Primary: Natural logarithm ln

shifted (2nd / f / SHIFT layer) (press #6 first): ex — e^x — exponential

alpha layer (press #13 first): S — Variable S (store with STO/ →, recall by name)

Keyboard: n

#37 4 Digits & number entry

Primary: Digit 4

shifted (2nd / f / SHIFT layer) (press #6 first): L4 — LIST menu

alpha layer (press #13 first): T — Variable T (store with STO/ →, recall by name)

Keyboard: 4

#38 5 Digits & number entry

Primary: Digit 5

shifted (2nd / f / SHIFT layer) (press #6 first): L5 — LIST menu

alpha layer (press #13 first): U — Variable U (store with STO/ →, recall by name)

Keyboard: 5

#39 6 Digits & number entry

Primary: Digit 6

shifted (2nd / f / SHIFT layer) (press #6 first): L6 — LIST menu

alpha layer (press #13 first): V — Variable V (store with STO/ →, recall by name)

Keyboard: 6

#40 – Operators & execute

Primary: Subtraction –

shifted (2nd / f / SHIFT layer) (press #6 first):] — Types "]"

alpha layer (press #13 first): W — Variable W (store with STO/ →, recall by name)

Keyboard: -

#41 STO▶ Memory, variables & registers

Primary: STO▶ (→) — store into a variable

shifted (2nd / f / SHIFT layer) (press #6 first): RCL — Recall a stored variable

alpha layer (press #13 first): X — Variable X (store with STO/ →, recall by name)

#42 1 Digits & number entry

Primary: Digit 1

shifted (2nd / f / SHIFT layer) (press #6 first): L1 — LIST menu

alpha layer (press #13 first): Y — Variable Y (store with STO/ →, recall by name)

Keyboard: 1

#43 2 Digits & number entry

Primary: Digit 2

shifted (2nd / f / SHIFT layer) (press #6 first): L2 — LIST menu

alpha layer (press #13 first): Z — Variable Z (store with STO/ →, recall by name)

Keyboard: 2

#44 3 Digits & number entry

Primary: Digit 3

shifted (2nd / f / SHIFT layer) (press #6 first): L3 — LIST menu

alpha layer (press #13 first): θ — Variable θ (store with STO/ →, recall by name)

Keyboard: 3

#45 + Operators & execute

Primary: Addition +

shifted (2nd / f / SHIFT layer) (press #6 first): MEM — RESET — return to factory defaults

Keyboard: +

#46 ON `Clear, delete & power`

Primary: ON — power on

shifted (2nd / f / SHIFT layer) (press #6 first): OFF — OFF — power off (state is lost; press ON to restart)

#47 0 `Digits & number entry`

Primary: Digit 0

shifted (2nd / f / SHIFT layer) (press #6 first): CATALOG — CATALOG — list of all commands

Keyboard: `0`

#48 . `Digits & number entry`

Primary: Decimal point. Pressing it first gives "0."; a second point in the same number is ignored

shifted (2nd / f / SHIFT layer) (press #6 first): i — Imaginary unit i (complex numbers)

alpha layer (press #13 first): : — Colon : (multi-statement separator)

Keyboard: `.`

#49 (−) `Digits & number entry`

Primary: Negative sign (−) — makes the next number negative. Different from the subtraction key

shifted (2nd / f / SHIFT layer) (press #6 first): ANS — Ans — recall the last answer into the current expression

#50 ENTER Operators & execute

Primary: Execute / = — evaluate the line and show the result (EXE / ENTER / =)

shifted (2nd / f / SHIFT layer) (press #6 first): ENTRY — ENTRY — recall a previous entry

alpha layer (press #13 first): SOLVE — Solver app

Keyboard: Enter =

5 · Worked examples

20 examples, each run on this calculator's keys by the U-45 checker. Follow the chips in order.

Example 5.1 — Order of operations

Calculate 12 + 3 × 4. Press Reset first.

#42 1 → #43 2 → #45 + → #44 3 → #35 × → #37 4 → #50 ENTER

Display shows 24 **— expected 24. ✔ verified on the app.**

Multiplication is done before addition, so the calculator holds the + until the × part is finished.

Example 5.2 — Brackets first

Calculate (12 + 8) ÷ 4. Press Reset first.

#28 (→ #42 1 → #43 2 → #45 + → #33 8 → #29) → #30 ÷ → #37 4 → #50 ENTER

Display shows 5 **— expected 5. ✔ verified on the app.**

Brackets force the sum to be worked out before the division.

Example 5.3 — Two brackets multiplied

Calculate (3 + 4) × (9 − 2). Press Reset first.

#28 (→ #44 3 → #45 + → #37 4 → #29) → #35 × → #28 (→ #34 9 → #40 − → #43 2 → #29) → #50 ENTER

Display shows 49 **— expected 49. ✔ verified on the app.**

Both brackets are evaluated first, then multiplied.

Example 5.4 — Squares

Calculate 12^2. Press Reset first.

#42 1 → #43 2 → #26 x^2 → #50 ENTER

Display shows 144 **— expected 144. ✔ verified on the app.**

x^2 squares the value — quicker and safer than typing the number twice.

Example 5.5 — Square roots

Calculate √144. Press Reset first.

#6 2nd → #26 √ → #42 1 → #37 4 → #37 4 → #50 ENTER

Display shows 12 **— expected 12. ✔ verified on the app.**

The square-root key returns the principal (positive) root.

Example 5.6 — Powers

Calculate **2^10**. Press **Reset** first.

`#43` 2 → `#25` ^ → `#42` 1 → `#47` 0 → `#50` ENTER

Display shows `1024` — expected 1024. ✔ verified on the app.

The power key raises the first number to the second.

Example 5.7 — Reciprocals

Calculate **1 ÷ 4 (x^{-1})**. Press **Reset** first.

`#37` 4 → `#21` x^{-1} → `#50` ENTER

Display shows `0.25` — expected 0.25. ✔ verified on the app.

x^{-1} (1/x) gives the reciprocal — handy for resistors in parallel or rates.

Example 5.8 — Trigonometry (degrees)

Calculate **cos 0° (Deg mode)**. Press **Reset** first.

`#23` COS → `#47` 0 → `#29`) → `#50` ENTER

Display shows `1` — expected 1. ✔ verified on the app.

Trig functions use the current angle unit; these answers assume degrees (the start-up setting).

Example 5.9 — Common logarithm

Calculate **log 1000**. Press **Reset** first.

`#31` LOG → `#42` 1 → `#47` 0 → `#47` 0 → `#47` 0 → `#29`) → `#50` ENTER

Display shows `3` — expected 3. ✔ verified on the app.

log is base-10: it counts the zeros of a power of ten.

Example 5.10 — Natural logarithm

Calculate **ln 1**. Press **Reset** first.

`#36` LN → `#42` 1 → `#29`) → `#50` ENTER

Display shows `0` — expected 0. ✔ verified on the app.

ln is the natural (base-e) logarithm.

Example 5.11 — Antilog 10^x

Calculate **10^2**. Press **Reset** first.

`#6` 2nd → `#31` 10^x → `#43` 2 → `#29`) → `#50` ENTER

Display shows `100` — expected 100. ✔ verified on the app.

10^x is the inverse of log.

Example 5.12 — Exponential e^x

Calculate e^1. Press Reset first.

#6 2nd → #36 e^x → #42 1 → #29) → #50 ENTER

Display shows 2.718281828 **— expected 2.71828183. ✔ verified on the app.**

e^x is the inverse of ln.

Example 5.13 — Negative numbers

Calculate −5 + 12. Press Reset first.

#49 (−) → #38 5 → #45 + → #42 1 → #43 2 → #50 ENTER

Display shows 7 **— expected 7. ✔ verified on the app.**

Use the sign key for a negative number — not the subtraction key.

Example 5.14 — Scientific notation

Calculate 6×10^2 × 3. Press Reset first.

#39 6 → #6 2nd → #27 EE → #43 2 → #35 × → #44 3 → #50 ENTER

Display shows 1800 **— expected 1800. ✔ verified on the app.**

The exponent key enters $\times 10^n$ without typing × 10 ^.

Example 5.15 — Area of a circle πr^2

Calculate $\pi \times 3^2$. Press Reset first.

#6 2nd → #25 π → #35 × → #44 3 → #26 x^2 → #50 ENTER

Display shows 28.27433388 **— expected 28.27433388. ✔ verified on the app.**

π is stored to full precision — never type 3.14.

Example 5.16 — Continue from the answer

Calculate 100 × 4 = , then Ans ÷ 2. Press Reset first.

#42 1 → #47 0 → #47 0 → #35 × → #37 4 → #50 ENTER →
#30 ÷ → #43 2 → #50 ENTER

Display shows 200 **— expected 200. ✔ verified on the app.**

Pressing an operator straight after a result continues from that answer (Ans).

Example 5.17 — Average of three numbers

Calculate (4 + 8 + 15) ÷ 3. Press Reset first.

#28 (→ #37 4 → #45 + → #33 8 → #45 + → #42 1 →
#38 5 → #29) → #30 ÷ → #44 3 → #50 ENTER

Display shows 9 **— expected 9. ✔ verified on the app.**

Add up the values, then divide by how many there are.

Example 5.18 — Mixed operations

Calculate 50 − 4 × 6 + 7. Press Reset first.

#38 5 → #47 0 → #40 − → #37 4 → #35 × → #39 6 → #45 + → #32 7 → #50 ENTER

Display shows 33 **— expected 33. ✔ verified on the app.**

Only the × part is done first; the − and + are then worked left to right.

Example 5.19 — Dividing twice

Calculate 360 ÷ 4 ÷ 9. Press Reset first.

#44 3 → #39 6 → #47 0 → #30 ÷ → #37 4 → #30 ÷ → #34 9 → #50 ENTER

Display shows 10 **— expected 10. ✔ verified on the app.**

Division is worked left to right: the first quotient is divided again.

Example 5.20 — Brackets inside a longer formula

Calculate ((3 + 5) × 4 − 2) ÷ 10. Press Reset first.

#28 (→ #28 (→ #44 3 → #45 + → #38 5 → #29) → #35 × → #37 4 → #40 − → #43 2 → #29) → #30 ÷ → #42 1 → #47 0 → #50 ENTER

Display shows 3 — expected 3. ✔ verified on the app.

The innermost brackets are worked out first, then the outer brackets, then the division.

6 · Tricks & hacks

Hack 1: History reuse

Scroll up in the home screen history and press ENTER/EXE to copy an old entry or answer.

Hack 2: Ans chaining

Start an entry with an operator to continue from the previous answer.

Hack 3: Store variables

Use STO▶ / → to store values into letters A–Z, then use them in expressions.

Hack 4: Table of values

Use the TABLE view to scan where a function changes sign before solving.

Hack 5: Window first

If a graph looks empty, adjust the WINDOW / zoom standard.

Hack 6: Edit instead of retyping

After a result, press ◀ or ▶ to bring the last expression back with the cursor in it; change one number and press = again.

Hack 7: Ans chains

Start a line with an operator (e.g. ×) and the calculator inserts Ans automatically — great for compound growth: 1.05 ×, then = = = ...

Hack 8: Exact vs decimal

Natural-display models keep fractions, √ and π exact. Press S⇔D to flip to decimal; press it again to return.

Hack 9: Close brackets for free

Trailing closing brackets can be left off; = closes them for you.

Hack 10: Implicit multiplication

Type 2π or 3(4+1) without ×. Implicit multiplication binds tighter than ÷ on textbook models — use brackets if in doubt.

Hack 11: Store constants

Store a value you reuse (e.g. g = 9.81) with STO and a letter; then type the letter in formulas.

Hack 12: Power chains

x^y chains right to left: 2^3^2 = 2^9 = 512, not 64. Use brackets: (2^3)^2 = 64.

Hack 13: Angles

Check the D / R / G indicator before trig. sin 30 = 0.5 only in degrees.

Hack 14: Memory sums

M+ / M− keep a running total separate from the calculation line; RCL M shows it.

Hack 15: Replay history

Use ▲ ▼ to scroll earlier lines; press = on an old line to re-run it.

Hack 16: Hidden functions on this model

70 extra functions sit on shifted layers. Press the layer key once, then the key: for example #6 then #1 → STAT PLOT; #6 then #2 → TBLSET; #6 then #3 → FORMAT; #6 then #4 → CALC.

Hack 17: Find any key by number

Use Appendix B (function finder): look up the function, read its key number, then find that number on the keypad. Numbers run left→right, top→bottom from #1 to #50.

Hack 18: Type faster on a computer

29 keys on this model have a keyboard shortcut (see chapter 7). Typing digits and operators is much faster than clicking.

Hack 19: Check before you trust

Estimate first (round each number) and compare with the display. If the answer is 10× or 100× off, a decimal point or percent key was missed.

Hack 20: Copy results out

Press Copy in the bar above the calculator to put the display text on the clipboard instead of retyping it.

Hack 21: Colour blind or low vision?

Hi-Contrast colours use black/white/yellow/blue/green/red with black or white text for maximum contrast; key numbers stay on in both colour schemes.

7 · Keyboard shortcuts

With the calculator open on a computer, these physical keys press the numbered keys below (Ctrl/Alt/⌘ combinations are left to the browser). Shifted functions are reached by clicking the shift key first.

Keyboard	#	Key	Does
Backspace Delete	#8	DEL	DEL — delete the character at the cursor
Escape	#20	CLEAR	CLEAR — clear the entry line / screen
i	#21	x^{-1}	Reciprocal x^{-1} (1/x)
s	#22	SIN	Sine (uses the current angle unit)
c	#23	COS	Cosine (current angle unit)
t	#24	TAN	Tangent (current angle unit)
^	#25	^	Power x^y / ^ — raise to any power. Chains right-to-left (2^3^2 = 2^9 = 512)
q	#26	x^2	Square x^2
,	#27	,	Comma / argument separator (between function arguments, list items or coordinates)
(	#28	(	Opening bracket (
)	#29	)	Closing bracket). Closing brackets at the end of a line may be omitted
/	#30	÷	Division ÷
l	#31	LOG	Common logarithm $\log_{10}$
7	#32	7	Digit 7
8	#33	8	Digit 8
9	#34	9	Digit 9
* x	#35	×	Multiplication ×
n	#36	LN	Natural logarithm ln
4	#37	4	Digit 4
5	#38	5	Digit 5

6	#39	6	Digit 6
-	#40	–	Subtraction –
1	#42	1	Digit 1
2	#43	2	Digit 2
3	#44	3	Digit 3
+	#45	+	Addition +
0	#47	0	Digit 0
.	#48	.	Decimal point. Pressing it first gives “0.”; a second point in the same number is ignored
Enter =	#50	ENTER	Execute / = — evaluate the line and show the result (EXE / ENTER / =)

App shortcuts

“# On / # Off”	show / hide key numbers
“Classic / Hi-Contrast”	switch key colours
“Copy”	copy display text
“Fit / 1:1”	scale to screen / design size
“Reset”	restart this calculator
Jump list	switch to another of the 45 calculators

8 · Common mistakes

Mistake 1: (–) vs –

Use the negative key for negative numbers.

Mistake 2: Radian mode

Graphs of sin x in degrees look flat over –10..10; switch to radians for standard trig windows.

Mistake 3: Implicit multiplication

2/3x may be read as 2/(3x) depending on the model; use brackets.

Mistake 4: Using – instead of (–)

The negative key (–) and the subtraction key are different; "– 5" at the start uses Ans – 5.

Mistake 5: Wrong angle unit

If sin 30 gives –0.988, the calculator is in radians. Switch to degrees in SETUP.

Mistake 6: -3^2 vs $(-3)^2$

$-3^2 = -9$ because the square is applied first. Type $(-3)^2$ to get 9.

Mistake 7: Division precedence

1 ÷ 2π may be read as 1 ÷ (2π) on textbook models. Type 1 ÷ (2 × π) explicitly to be sure.

Mistake 8: Forgetting SHIFT is one-shot

SHIFT applies only to the next key. If the S indicator is still lit, press SHIFT again to cancel.

Mistake 9: Clearing memory by accident

AC clears the line but not memory; CLR (SHIFT 9) can clear memory and variables.

Mistake 10: Forgetting Reset between problems

Old values (memory, stack, open brackets, worksheet values) can leak into the next calculation. Press Reset or clear before starting a new problem.

Mistake 11: Pressing the wrong layer

A shift / 2nd indicator that is still lit will turn the next key into its shifted function. Look at the display indicators before pressing.

Mistake 12: Minus vs negative

The (−) negative key and the − subtraction key are different. Use (−) to start a negative number, − between two numbers.

9 · Practice drills

Set A — Calculation drills (107)

Work each one on calculator #2, then check your keys and the answer in Appendix A. Every answer was produced by this calculator.

1. 25 + 5 × 6 = _____
2. 7 + 8 × 9 = _____
3. 150 + 12 × 11 = _____
4. 3 + 14 × 15 = _____
5. 48 + 6 × 7 = _____
6. 1000 + 25 × 4 = _____
7. 19 + 21 × 3 = _____
8. 64 + 8 × 8 = _____
9. (45 + 15) ÷ 6 = _____
10. (9 + 7) ÷ 2 = _____
11. (100 + 60) ÷ 8 = _____
12. (33 + 17) ÷ 5 = _____
13. (1 + 2) ÷ 4 = _____
14. (250 + 350) ÷ 12 = _____
15. (5 + 6) × (10 − 3) = _____
16. (12 + 3) × (8 − 5) = _____

17. (1.5 + 2.5) × (7 − 4) = _____

18. (20 + 30) × (15 − 5) = _____

19. (9 + 1) × (100 − 1) = _____

20. (11 + 13) × (17 − 9) = _____

21. 25^2 = _____

22. 7^2 = _____

23. 1.5^2 = _____

24. 19^2 = _____

25. 33^2 = _____

26. 0.5^2 = _____

27. 125^2 = _____

28. 64^2 = _____

29. $\sqrt{625}$ = _____

30. $\sqrt{2}$ = _____

31. $\sqrt{50}$ = _____

32. $\sqrt{1024}$ = _____

33. $\sqrt{3}$ = _____

34. $\sqrt{9801}$ = _____

35. 3^5 = _____

36. 1.05^3 = _____

37. 5^4 = _____

38. 7^3 = _____

39. 1.5^2 = _____

40. 10^6 = _____

41. 2^0.5 = _____

42. 1 ÷ 8 (x^{-1}) = _____

43. 1 ÷ 0.25 (x^{-1}) = _____

44. 1 ÷ 16 (x^{-1}) = _____

45. 1 ÷ 40 (x^{-1}) = _____

46. 1 ÷ 0.2 (x^{-1}) = _____

47. log 100 = _____

48. log 10000 = _____

49. ln 10 = _____

50. ln 2 = _____

51. 10^3 = _____

52. e^2 = _____

53. −20 + 7 = _____

54. −3 + 3 = _____

55. −100 + 250 = _____

56. −8 + 1 = _____

57. 2×10^3 × 4 = _____

58. 1.5×10^4 × 2 = _____

59. 3×10^5 × 3 = _____

60. 4.5×10^2 × 2 = _____

61. $\pi \times 10^2$ = _____

62. $\pi \times 2.5^2$ = _____

63. $\pi \times 7^2$ = _____

64. $\pi \times 1.2^2$ = _____

65. 15 × 6 = , then Ans ÷ 9 = _____

66. 7.5 × 8 = , then Ans ÷ 3 = _____

67. 250 × 12 = , then Ans ÷ 100 = _____

68. (12 + 15 + 18) ÷ 3 = _____

69. (70 + 85 + 91) ÷ 3 = _____

70. (2.5 + 3.5 + 6) ÷ 3 = _____

71. (100 + 200 + 600) ÷ 3 = _____

72. 100 − 3 × 12 + 9 = _____

73. 18 − 2 × 5 + 40 = _____

74. 7.5 − 1.5 × 3 + 2 = _____

75. 64 − 8 × 4 + 10 = _____

76. 300 − 25 × 4 + 55 = _____

77. $9 - 3 \times 3 + 1 =$ _____

78. $45 - 5 \times 5 + 15 =$ _____

79. $1000 \div 8 \div 5 =$ _____

80. $96 \div 2 \div 3 =$ _____

81. $7200 \div 60 \div 60 =$ _____

82. $500 \div 4 \div 25 =$ _____

83. $81 \div 3 \div 3 =$ _____

84. $((10 + 2) \times 3 - 6) \div 6 =$ _____

85. $((1.5 + 2.5) \times 6 - 4) \div 5 =$ _____

86. $((20 + 30) \times 2 - 25) \div 3 =$ _____

87. $((7 + 8) \times 9 - 15) \div 12 =$ _____

88. $12 + 3 \times 4 =$ _____

89. $(12 + 8) \div 4 =$ _____

90. $(3 + 4) \times (9 - 2) =$ _____

91. $12^2 =$ _____

92. $\sqrt{144} =$ _____

93. 2^10 = _____

94. $1 \div 4\ (x^{-1}) =$ _____

95. cos 0° (Deg mode) = _____

96. log 1000 = _____

97. ln 1 = _____

98. 10^2 = _____

99. e^1 = _____

100. −5 + 12 = _____

101. 6×10^2 × 3 = _____

102. π × 3^2 = _____

103. 100 × 4 = , then Ans ÷ 2 = _____

104. (4 + 8 + 15) ÷ 3 = _____

105. 50 − 4 × 6 + 7 = _____

106. 360 ÷ 4 ÷ 9 = _____

107. ((3 + 5) × 4 − 2) ÷ 10 = _____

Set B — Key-finding drills (110)

Without looking at the table, find each key on the picture. Answers in Appendix A.

1. Which key is “Y=” (Y=)? _____
2. Which key is “WINDOW” (WINDOW)? _____
3. Which key is “ZOOM” (ZOOM)? _____
4. Which key is “TRACE” (TRACE)? _____
5. Which key is “GRAPH” (GRAPH)? _____
6. Which key is “2nd” (SHIFT / 2nd)? _____
7. Which key is “MODE” (MODE)? _____

8. **Which key is “DEL” (DEL)?** _____

9. **Which key is “▲” (Cursor ▲)?** _____

10. **Which key is “▼” (Cursor ▼)?** _____

11. **Which key is “◀” (Cursor ◀)?** _____

12. **Which key is “▶” (Cursor ▶)?** _____

13. **Which key is “ALPHA” (ALPHA)?** _____

14. **Which key is “X,T,θ,n” (The graphing variable)?** _____

15. **Which key is “STAT” (STAT)?** _____

16. **Which key is “MATH” (MATH menu)?** _____

17. **Which key is “APPS” (APPS)?** _____

18. **Which key is “PRGM” (PRGM)?** _____

19. **Which key is “VARS” (VARS)?** _____

20. **Which key is “CLEAR” (CLEAR)?** _____

21. **Which key is “x^{-1}” (Reciprocal x^{-1})?** _____

22. **Which key is “SIN” (Sine)?** _____

23. **Which key is “COS” (Cosine)?** _____

24. **Which key is “TAN” (Tangent)?** _____

25. **Which key is “^” (Power x^y / ^)?** _____

26. **Which key is “x^2” (Square x^2)?** _____

27. **Which key is “,” (Comma / argument separator)?** _____

28. Which key is “(” (Opening bracket)? _____

29. Which key is “)” (Closing bracket). Closing brackets at the end of a line may be omitted)? _____

30. Which key is “÷” (Division ÷)? _____

31. Which key is “LOG” (Common logarithm $\log_{10}$)? _____

32. Which key is “×” (Multiplication ×)? _____

33. Which key is “LN” (Natural logarithm ln)? _____

34. Which key is “−” (Subtraction −)? _____

35. Which key is “STO▶” (STO▶)? _____

36. Which key is “+” (Addition +)? _____

37. Which key is “ON” (ON)? _____

38. Which key is “.” (Decimal point. Pressing it first gives “0.”; a second point in the same number is ignored)? _____

39. Which key is “(−)” (Negative sign)? _____

40. Which key is “ENTER” (Execute / =)? _____

41. How do you reach “STAT PLOT”? _____

42. How do you reach “TBLSET”? _____

43. How do you reach “FORMAT”? _____

44. How do you reach “CALC”? _____

45. How do you reach “TABLE”? _____

46. How do you reach “QUIT”? _____

47. How do you reach “INS”? _____

48. How do you reach “A-LOCK”? _____

49. How do you reach “LINK”? _____

50. How do you reach “LIST”? _____

51. How do you reach “TEST”? _____

52. How do you reach “A”? _____

53. How do you reach “ANGLE”? _____

54. How do you reach “B”? _____

55. How do you reach “DRAW”? _____

56. How do you reach “C”? _____

57. How do you reach “DISTR”? _____

58. How do you reach “MATRIX”? _____

59. How do you reach “D”? _____

60. How do you reach “SIN^{-1}”? _____

61. How do you reach “E”? _____

62. How do you reach “COS^{-1}”? _____

63. How do you reach “F”? _____

64. How do you reach “TAN^{-1}”? _____

65. How do you reach “G”? _____

66. How do you reach “π”? _____

67. How do you reach “H”? _____

68. How do you reach “√”? _____

69. How do you reach “I”? _____

70. How do you reach “EE”? _____

71. How do you reach “J”? _____

72. How do you reach “{”? _____

73. How do you reach “K”? _____

74. How do you reach “}”? _____

75. How do you reach “L”? _____

76. How do you reach “e”? _____

77. How do you reach “M”? _____

78. How do you reach “10x”? _____

79. How do you reach “N”? _____

80. How do you reach “O”? _____

81. How do you reach “P”? _____

82. How do you reach “Q”? _____

83. How do you reach “[”? _____

84. How do you reach “R”? _____

85. How do you reach “ex”? _____

86. How do you reach “S”? _____

87. How do you reach “L4”? _____

88. How do you reach “T”? _____

89. How do you reach “L5”? _____

90. How do you reach “U”? _____

91. How do you reach “L6”? _____

92. How do you reach “V”? _____

93. How do you reach “]”? _____

94. How do you reach “W”? _____

95. How do you reach “RCL”? _____

96. How do you reach “X”? _____

97. How do you reach “L1”? _____

98. How do you reach “Y”? _____

99. How do you reach “L2”? _____

100. How do you reach “Z”? _____

101. How do you reach “L3”? _____

102. How do you reach “θ”? _____

103. How do you reach “MEM”? _____

104. How do you reach “OFF”? _____

105. How do you reach “CATALOG”? _____

106. How do you reach “i”? _____

107. How do you reach ":"? _____

108. How do you reach "ANS"? _____

109. How do you reach "ENTRY"? _____

110. How do you reach "SOLVE"? _____

Appendix A — Drill answers

Set A

1. **55** (display `55`)

   `#43` 2 → `#38` 5 → `#45` + → `#38` 5 → `#35` × → `#39` 6 → `#50` ENTER

2. **79** (display `79`)

   `#32` 7 → `#45` + → `#33` 8 → `#35` × → `#34` 9 → `#50` ENTER

3. **282** (display `282`)

   `#42` 1 → `#38` 5 → `#47` 0 → `#45` + → `#42` 1 → `#43` 2 → `#35` × → `#42` 1 → `#42` 1 → `#50` ENTER

4. **213** (display `213`)

   `#44` 3 → `#45` + → `#42` 1 → `#37` 4 → `#35` × → `#42` 1 → `#38` 5 → `#50` ENTER

5. **90** (display `90`)

   `#37` 4 → `#33` 8 → `#45` + → `#39` 6 → `#35` × → `#32` 7 → `#50` ENTER

6. **1100** (display `1100`)

   `#42` 1 → `#47` 0 → `#47` 0 → `#47` 0 → `#45` + → `#43` 2 → `#38` 5 → `#35` × → `#37` 4 → `#50` ENTER

7. 82 (display `82`)

`#42` 1 → `#34` 9 → `#45` + → `#43` 2 → `#42` 1 → `#35` × → `#44` 3 → `#50` ENTER

8. 128 (display `128`)

`#39` 6 → `#37` 4 → `#45` + → `#33` 8 → `#35` × → `#33` 8 → `#50` ENTER

9. 10 (display `10`)

`#28` (→ `#37` 4 → `#38` 5 → `#45` + → `#42` 1 → `#38` 5 → `#29`) → `#30` ÷ → `#39` 6 → `#50` ENTER

10. 8 (display `8`)

`#28` (→ `#34` 9 → `#45` + → `#32` 7 → `#29`) → `#30` ÷ → `#43` 2 → `#50` ENTER

11. 20 (display `20`)

`#28` (→ `#42` 1 → `#47` 0 → `#47` 0 → `#45` + → `#39` 6 → `#47` 0 → `#29`) → `#30` ÷ → `#33` 8 → `#50` ENTER

12. 10 (display `10`)

`#28` (→ `#44` 3 → `#44` 3 → `#45` + → `#42` 1 → `#32` 7 → `#29`) → `#30` ÷ → `#38` 5 → `#50` ENTER

13. 0.75 (display `0.75`)

`#28` (→ `#42` 1 → `#45` + → `#43` 2 → `#29`) → `#30` ÷ → `#37` 4 → `#50` ENTER

14. 50 (display 50)

#28 (→ #43 2 → #38 5 → #47 0 → #45 + → #44 3 → #38 5 → #47 0 → #29) → #30 ÷ → #42 1 → #43 2 → #50 ENTER

15. 77 (display 77)

#28 (→ #38 5 → #45 + → #39 6 → #29) → #35 × → #28 (→ #42 1 → #47 0 → #40 − → #44 3 → #29) → #50 ENTER

16. 45 (display 45)

#28 (→ #42 1 → #43 2 → #45 + → #44 3 → #29) → #35 × → #28 (→ #33 8 → #40 − → #38 5 → #29) → #50 ENTER

17. 12 (display 12)

#28 (→ #42 1 → #48 . → #38 5 → #45 + → #43 2 → #48 . → #38 5 → #29) → #35 × → #28 (→ #32 7 → #40 − → #37 4 → #29) → #50 ENTER

18. 500 (display 500)

#28 (→ #43 2 → #47 0 → #45 + → #44 3 → #47 0 → #29) → #35 × → #28 (→ #42 1 → #38 5 → #40 − → #38 5 → #29) → #50 ENTER

19. 990 (display 990)

#28 (→ #34 9 → #45 + → #42 1 → #29) → #35 × →
#28 (→ #42 1 → #47 0 → #47 0 → #40 − → #42 1 →
#29) → #50 ENTER

20. 192 (display 192)

#28 (→ #42 1 → #42 1 → #45 + → #42 1 → #44 3 →
#29) → #35 × → #28 (→ #42 1 → #32 7 → #40 − →
#34 9 → #29) → #50 ENTER

21. 625 (display 625)

#43 2 → #38 5 → #26 x^2 → #50 ENTER

22. 49 (display 49)

#32 7 → #26 x^2 → #50 ENTER

23. 2.25 (display 2.25)

#42 1 → #48 . → #38 5 → #26 x^2 → #50 ENTER

24. 361 (display 361)

#42 1 → #34 9 → #26 x^2 → #50 ENTER

25. 1089 (display 1089)

#44 3 → #44 3 → #26 x^2 → #50 ENTER

26. 0.25 (display 0.25)

#47 0 → #48 . → #38 5 → #26 x^2 → #50 ENTER

27. 15625 (display `15625`)

`#42` 1 → `#43` 2 → `#38` 5 → `#26` x^2 → `#50` ENTER

28. 4096 (display `4096`)

`#39` 6 → `#37` 4 → `#26` x^2 → `#50` ENTER

29. 25 (display `25`)

`#6` 2nd → `#26` √ → `#39` 6 → `#43` 2 → `#38` 5 → `#50` ENTER

30. 1.41421356 (display `1.414213562`)

`#6` 2nd → `#26` √ → `#43` 2 → `#50` ENTER

31. 7.07106781 (display `7.071067812`)

`#6` 2nd → `#26` √ → `#38` 5 → `#47` 0 → `#50` ENTER

32. 32 (display `32`)

`#6` 2nd → `#26` √ → `#42` 1 → `#47` 0 → `#43` 2 → `#37` 4 → `#50` ENTER

33. 1.73205081 (display `1.732050808`)

`#6` 2nd → `#26` √ → `#44` 3 → `#50` ENTER

34. 99 (display `99`)

`#6` 2nd → `#26` √ → `#34` 9 → `#33` 8 → `#47` 0 → `#42` 1 → `#50` ENTER

35. 243 (display `243`)

`#44` 3 → `#25` ^ → `#38` 5 → `#50` ENTER

36. 1.157625 (display `1.157625`)

`#42` 1 → `#48` . → `#47` 0 → `#38` 5 → `#25` ^ → `#44` 3 → `#50` ENTER

37. 625 (display `625`)

`#38` 5 → `#25` ^ → `#37` 4 → `#50` ENTER

38. 343 (display `343`)

`#32` 7 → `#25` ^ → `#44` 3 → `#50` ENTER

39. 2.25 (display `2.25`)

`#42` 1 → `#48` . → `#38` 5 → `#25` ^ → `#43` 2 → `#50` ENTER

40. 1000000.000 (display `1000000`)

`#42` 1 → `#47` 0 → `#25` ^ → `#39` 6 → `#50` ENTER

41. 1.41421356 (display `1.414213562`)

`#43` 2 → `#25` ^ → `#47` 0 → `#48` . → `#38` 5 → `#50` ENTER

42. 0.125 (display `0.125`)

`#33` 8 → `#21` x^{-1} → `#50` ENTER

43. 4 (display `4`)

`#47` 0 → `#48` . → `#43` 2 → `#38` 5 → `#21` x^{-1} → `#50` ENTER

44. 0.0625 (display `0.0625`)

`#42` 1 → `#39` 6 → `#21` x^{-1} → `#50` ENTER

45. 0.025 (display `0.025`)

`#37` 4 → `#47` 0 → `#21` x^{-1} → `#50` ENTER

46. 5 (display `5`)

`#47` 0 → `#48` . → `#43` 2 → `#21` x^{-1} → `#50` ENTER

47. 2 (display `2`)

`#31` LOG → `#42` 1 → `#47` 0 → `#47` 0 → `#29`) → `#50` ENTER

48. 4 (display `4`)

`#31` LOG → `#42` 1 → `#47` 0 → `#47` 0 → `#47` 0 → `#47` 0 → `#29`) → `#50` ENTER

49. 2.30258509 (display `2.302585093`)

`#36` LN → `#42` 1 → `#47` 0 → `#29`) → `#50` ENTER

50. 0.69314718 (display `0.6931471806`)

`#36` LN → `#43` 2 → `#29`) → `#50` ENTER

51. 1000 (display `1000`)

`#6` 2nd → `#31` 10^x → `#44` 3 → `#29`) → `#50` ENTER

52. 7.3890561 (display `7.389056099`)

`#6` 2nd → `#36` e^x → `#43` 2 → `#29`) → `#50` ENTER

53. -13 (display `-13`)

`#49` (–) → `#43` 2 → `#47` 0 → `#45` + → `#32` 7 → `#50` ENTER

54. 0 (display `0`)

`#49` (–) → `#44` 3 → `#45` + → `#44` 3 → `#50` ENTER

55. 150 (display `150`)

#49 (−) → #42 1 → #47 0 → #47 0 → #45 + → #43 2 → #38 5 → #47 0 → #50 ENTER

56. -7 (display `-7`)

#49 (−) → #33 8 → #45 + → #42 1 → #50 ENTER

57. 8000 (display `8000`)

#43 2 → #6 2nd → #27 EE → #44 3 → #35 × → #37 4 → #50 ENTER

58. 30000 (display `30000`)

#42 1 → #48 . → #38 5 → #6 2nd → #27 EE → #37 4 → #35 × → #43 2 → #50 ENTER

59. 900000 (display `900000`)

#44 3 → #6 2nd → #27 EE → #38 5 → #35 × → #44 3 → #50 ENTER

60. 900 (display `900`)

#37 4 → #48 . → #38 5 → #6 2nd → #27 EE → #43 2 → #35 × → #43 2 → #50 ENTER

61. 314.15926536 (display `314.1592654`)

#6 2nd → #25 π → #35 × → #42 1 → #47 0 → #26 x^2 → #50 ENTER

62. 19.63495408 (display `19.63495408`)

#6 2nd → #25 π → #35 × → #43 2 → #48 . → #38 5 → #26 x^2 → #50 ENTER

63. 153.93804003 (display `153.93804`)

#6 2nd → #25 π → #35 × → #32 7 → #26 x^2 → #50 ENTER

64. 4.52389342 (display `4.523893421`)

#6 2nd → #25 π → #35 × → #42 1 → #48 . → #43 2 → #26 x^2 → #50 ENTER

65. 10 (display `10`)

#42 1 → #38 5 → #35 × → #39 6 → #50 ENTER → #30 ÷ → #34 9 → #50 ENTER

66. 20 (display `20`)

#32 7 → #48 . → #38 5 → #35 × → #33 8 → #50 ENTER → #30 ÷ → #44 3 → #50 ENTER

67. 30 (display `30`)

#43 2 → #38 5 → #47 0 → #35 × → #42 1 → #43 2 → #50 ENTER → #30 ÷ → #42 1 → #47 0 → #47 0 → #50 ENTER

68. 15 (display 15)

#28 (→ #42 1 → #43 2 → #45 + → #42 1 → #38 5 → #45 + → #42 1 → #33 8 → #29) → #30 ÷ → #44 3 → #50 ENTER

69. 82 (display 82)

#28 (→ #32 7 → #47 0 → #45 + → #33 8 → #38 5 → #45 + → #34 9 → #42 1 → #29) → #30 ÷ → #44 3 → #50 ENTER

70. 4 (display 4)

#28 (→ #43 2 → #48 . → #38 5 → #45 + → #44 3 → #48 . → #38 5 → #45 + → #39 6 → #29) → #30 ÷ → #44 3 → #50 ENTER

71. 300 (display 300)

#28 (→ #42 1 → #47 0 → #47 0 → #45 + → #43 2 → #47 0 → #47 0 → #45 + → #39 6 → #47 0 → #47 0 → #29) → #30 ÷ → #44 3 → #50 ENTER

72. 73 (display 73)

#42 1 → #47 0 → #47 0 → #40 − → #44 3 → #35 × → #42 1 → #43 2 → #45 + → #34 9 → #50 ENTER

73. 48 (display 48)



74. 5 (display 5)



75. 42 (display 42)



76. 255 (display 255)



77. 1 (display 1)



78. 35 (display 35)



79. 25 (display 25)



80. 16 (display 16)

#34 9 → #39 6 → #30 ÷ → #43 2 → #30 ÷ → #44 3 → #50 ENTER

81. 2 (display 2)

#32 7 → #43 2 → #47 0 → #47 0 → #30 ÷ → #39 6 → #47 0 → #30 ÷ → #39 6 → #47 0 → #50 ENTER

82. 5 (display 5)

#38 5 → #47 0 → #47 0 → #30 ÷ → #37 4 → #30 ÷ → #43 2 → #38 5 → #50 ENTER

83. 9 (display 9)

#33 8 → #42 1 → #30 ÷ → #44 3 → #30 ÷ → #44 3 → #50 ENTER

84. 5 (display 5)

#28 (→ #28 (→ #42 1 → #47 0 → #45 + → #43 2 → #29) → #35 × → #44 3 → #40 − → #39 6 → #29) → #30 ÷ → #39 6 → #50 ENTER

85. 4 (display 4)

#28 (→ #28 (→ #42 1 → #48 . → #38 5 → #45 + → #43 2 → #48 . → #38 5 → #29) → #35 × → #39 6 → #40 − → #37 4 → #29) → #30 ÷ → #38 5 → #50 ENTER

86. 25 (display `25`)

`#28` (→ `#28` (→ `#43` 2 → `#47` 0 → `#45` + → `#44` 3 → `#47` 0 → `#29`) → `#35` × → `#43` 2 → `#40` − → `#43` 2 → `#38` 5 → `#29`) → `#30` ÷ → `#44` 3 → `#50` ENTER

87. 10 (display `10`)

`#28` (→ `#28` (→ `#32` 7 → `#45` + → `#33` 8 → `#29`) → `#35` × → `#34` 9 → `#40` − → `#42` 1 → `#38` 5 → `#29`) → `#30` ÷ → `#42` 1 → `#43` 2 → `#50` ENTER

88. 24 (display `24`)

`#42` 1 → `#43` 2 → `#45` + → `#44` 3 → `#35` × → `#37` 4 → `#50` ENTER

89. 5 (display `5`)

`#28` (→ `#42` 1 → `#43` 2 → `#45` + → `#33` 8 → `#29`) → `#30` ÷ → `#37` 4 → `#50` ENTER

90. 49 (display `49`)

`#28` (→ `#44` 3 → `#45` + → `#37` 4 → `#29`) → `#35` × → `#28` (→ `#34` 9 → `#40` − → `#43` 2 → `#29`) → `#50` ENTER

91. 144 (display `144`)

`#42` 1 → `#43` 2 → `#26` x^2 → `#50` ENTER

92. 12 (display `12`)

`#6` 2nd → `#26` $\sqrt{}$ → `#42` 1 → `#37` 4 → `#37` 4 → `#50` ENTER

93. **1024** (display `1024`)

#43 2 → #25 ^ → #42 1 → #47 0 → #50 ENTER

94. **0.25** (display `0.25`)

#37 4 → #21 x^{-1} → #50 ENTER

95. **1** (display `1`)

#23 COS → #47 0 → #29) → #50 ENTER

96. **3** (display `3`)

#31 LOG → #42 1 → #47 0 → #47 0 → #47 0 → #29) → #50 ENTER

97. **0** (display `0`)

#36 LN → #42 1 → #29) → #50 ENTER

98. **100** (display `100`)

#6 2nd → #31 10^x → #43 2 → #29) → #50 ENTER

99. **2.71828183** (display `2.718281828`)

#6 2nd → #36 e^x → #42 1 → #29) → #50 ENTER

100. **7** (display `7`)

#49 (–) → #38 5 → #45 + → #42 1 → #43 2 → #50 ENTER

101. **1800** (display `1800`)

#39 6 → #6 2nd → #27 EE → #43 2 → #35 × → #44 3 → #50 ENTER

102. 28.27433388 (display `28.27433388`)

`#6` 2nd → `#25` π → `#35` × → `#44` 3 → `#26` x^2 → `#50` ENTER

103. 200 (display `200`)

`#42` 1 → `#47` 0 → `#47` 0 → `#35` × → `#37` 4 → `#50` ENTER → `#30` ÷ → `#43` 2 → `#50` ENTER

104. 9 (display `9`)

`#28` (→ `#37` 4 → `#45` + → `#33` 8 → `#45` + → `#42` 1 → `#38` 5 → `#29`) → `#30` ÷ → `#44` 3 → `#50` ENTER

105. 33 (display `33`)

`#38` 5 → `#47` 0 → `#40` − → `#37` 4 → `#35` × → `#39` 6 → `#45` + → `#32` 7 → `#50` ENTER

106. 10 (display `10`)

`#44` 3 → `#39` 6 → `#47` 0 → `#30` ÷ → `#37` 4 → `#30` ÷ → `#34` 9 → `#50` ENTER

107. 3 (display `3`)

`#28` (→ `#28` (→ `#44` 3 → `#45` + → `#38` 5 → `#29`) → `#35` × → `#37` 4 → `#40` − → `#43` 2 → `#29`) → `#30` ÷ → `#42` 1 → `#47` 0 → `#50` ENTER

Set B

1. #1
2. #2
3. #3
4. #4
5. #5
6. #6
7. #7
8. #8
9. #9
10. #10
11. #11
12. #12
13. #13
14. #14
15. #15
16. #16
17. #17
18. #18
19. #19
20. #20
21. #21
22. #22
23. #23
24. #24
25. #25
26. #26
27. #27
28. #28
29. #29
30. #30
31. #31
32. #35
33. #36
34. #40
35. #41
36. #45
37. #46
38. #48
39. #49
40. #50
41. #6 2nd then #1 Y=
42. #6 2nd then #2 WINDOW
43. #6 2nd then #3 ZOOM
44. #6 2nd then #4 TRACE
45. #6 2nd then #5 GRAPH
46. #6 2nd then #7 MODE
47. #6 2nd then #8 DEL
48. #6 2nd then #13 ALPHA
49. #6 2nd then #14 X,T,θ,n
50. #6 2nd then #15 STAT
51. #6 2nd then #16 MATH
52. #13 ALPHA then #16 MATH
53. #6 2nd then #17 APPS
54. #13 ALPHA then #17 APPS
55. #6 2nd then #18 PRGM
56. #13 ALPHA then #18 PRGM
57. #6 2nd then #19 VARS
58. #6 2nd then #21 x^{-1}
59. #13 ALPHA then #21 x^{-1}
60. #6 2nd then #22 SIN
61. #13 ALPHA then #22 SIN
62. #6 2nd then #23 COS
63. #13 ALPHA then #23 COS
64. #6 2nd then #24 TAN
65. #13 ALPHA then #24 TAN
66. #6 2nd then #25 ^
67. #13 ALPHA then #25 ^
68. #6 2nd then #26 x^2
69. #13 ALPHA then #26 x^2
70. #6 2nd then #27 ,
71. #13 ALPHA then #27 ,
72. #6 2nd then #28 (
73. #13 ALPHA then #28 (
74. #6 2nd then #29)
75. #13 ALPHA then #29)
76. #6 2nd then #30 ÷
77. #13 ALPHA then #30 ÷
78. #6 2nd then #31 LOG
79. #13 ALPHA then #31 LOG
80. #13 ALPHA then #32 7
81. #13 ALPHA then #33 8
82. #13 ALPHA then #34 9

83. #6 2nd then #35 ×
84. #13 ALPHA then #35 ×
85. #6 2nd then #36 LN
86. #13 ALPHA then #36 LN
87. #6 2nd then #37 4
88. #13 ALPHA then #37 4
89. #6 2nd then #38 5
90. #13 ALPHA then #38 5
91. #6 2nd then #39 6
92. #13 ALPHA then #39 6
93. #6 2nd then #40 -
94. #13 ALPHA then #40 -
95. #6 2nd then #41 STO▸
96. #13 ALPHA then #41 STO▸
97. #6 2nd then #42 1
98. #13 ALPHA then #42 1
99. #6 2nd then #43 2
100. #13 ALPHA then #43 2
101. #6 2nd then #44 3
102. #13 ALPHA then #44 3
103. #6 2nd then #45 +
104. #6 2nd then #46 ON
105. #6 2nd then #47 0
106. #6 2nd then #48 .
107. #13 ALPHA then #48 .
108. #6 2nd then #49 (-)
109. #6 2nd then #50 ENTER
110. #13 ALPHA then #50 ENTER

Appendix B — Function finder (A–Z)

Function	Label	Keys
×10^x / EXP / EE — enter a power-of-ten exponent (scientific notation), e.g. 6.02 ×10^x 23	EE	#6 → #27
10^x — antilog	10x	#6 → #31
Addition +	+	#45
ALPHA — the next key types its letter/variable (red/green label)	ALPHA	#13
ALPHA — the next key types its letter/variable (red/green label)	A-LOCK	#6 → #13
Angle symbol ∠ (polar complex form)	ANGLE	#6 → #17
Ans — recall the last answer into the current expression	ANS	#6 → #49
APPS — applications	APPS	#17
CALC — graph analysis (zero, min, max, intersect)	CALC	#6 → #4
CATALOG — list of all commands	CATALOG	#6 → #47
CLEAR — clear the entry line / screen	CLEAR	#20
Closing bracket). Closing brackets at the end of a line may be omitted	)	#29
Colon : (multi-statement separator)	:	#13 → #48
Comma / argument separator (between function arguments, list items or coordinates)	,	#27
Common logarithm $\log_{10}$	LOG	#31
Cosine (current angle unit)	COS	#23
Cursor ▲ — up / previous history line / previous item	▲	#9
Cursor ▶ — move right in the line	▶	#12
Cursor ▼ — down / next history line / next item	▼	#10
Cursor ◀ — move left in the line	◀	#11

Decimal point. Pressing it first gives "0."; a second point in the same number is ignored	.	#48
DEL — delete the character at the cursor	DEL	#8
Digit 0	0	#47
Digit 1	1	#42
Digit 2	2	#43
Digit 3	3	#44
Digit 4	4	#37
Digit 5	5	#38
Digit 6	6	#39
Digit 7	7	#32
Digit 8	8	#33
Digit 9	9	#34
DISTR — probability distributions	DISTR	#6 → #19
Division ÷	÷	#30
DRAW menu	FORMAT	#6 → #3
DRAW menu	DRAW	#6 → #18
ENTRY — recall a previous entry	ENTRY	#6 → #50
Euler's number e (2.71828…)	e	#6 → #30
e^x — exponential	ex	#6 → #36
Execute / = — evaluate the line and show the result (EXE / ENTER / =)	ENTER	#50
GRAPH — draw the graph	GRAPH	#5
Imaginary unit i (complex numbers)	i	#6 → #48
INS — insert mode at the cursor	INS	#6 → #8

Inverse cosine $\cos^{-1}$	$\cos^{-1}$	#6 → #23
Inverse sine $\sin^{-1}$	$\sin^{-1}$	#6 → #22
Inverse tangent $\tan^{-1}$	$\tan^{-1}$	#6 → #24
LIST menu	LIST	#6 → #15
LIST menu	L4	#6 → #37
LIST menu	L5	#6 → #38
LIST menu	L6	#6 → #39
LIST menu	L1	#6 → #42
LIST menu	L2	#6 → #43
LIST menu	L3	#6 → #44
MATH menu — extra math functions	MATH	#16
MATRIX menu	MATRIX	#6 → #21
MODE — choose calculation mode	MODE	#7
Multiplication ×	×	#35
Natural logarithm ln	LN	#36
Negative sign (−) — makes the next number negative. Different from the subtraction key	(−)	#49
OFF — power off (state is lost; press ON to restart)	OFF	#6 → #46
ON — power on	ON	#46
Opening bracket (	(	#28

Power x^y / ^ — raise to any power. Chains right-to-left (2^3^2 = 2^9 = 512)	^	#25
PRGM — programs	PRGM	#18
QUIT — leave the current screen / app	QUIT	#6 → #7
Recall a stored variable	RCL	#6 → #41
Reciprocal x^{-1} (1/x)	x^{-1}	#21
RESET — return to factory defaults	MEM	#6 → #45
SHIFT / 2nd — the next key uses its upper (shift-coloured) function	2nd	#6
Shows the message "Link not available" (feature not provided in this app)	LINK	#6 → #14
Shows the message "TEST: type = < > with ALPHA" (feature not provided in this app)	TEST	#6 → #16
Sine (uses the current angle unit)	SIN	#22
Solver app	SOLVE	#13 → #50
Square root √	√	#6 → #26
Square x^2	x^2	#26
STAT	STAT	#15
STAT PLOT	STAT PLOT	#6 → #1
STO▶ (→) — store into a variable	STO▸	#41
Subtraction –	–	#40
TABLE — table of values	TABLE	#6 → #5
Tangent (current angle unit)	TAN	#24
TBLSET — table start and step	TBLSET	#6 → #2
The graphing variable (X / x,θ,t,n) for Y= functions, tables and solving	X,T,θ,n	#14
TRACE — move along a graph	TRACE	#4

Types "["	[	#6 → #35
Types "]"	]	#6 → #40
Types "{"	{	#6 → #28
Types "}"	}	#6 → #29
Variable A (store with STO/→, recall by name)	A	#13 → #16
Variable B (store with STO/→, recall by name)	B	#13 → #17
Variable C (store with STO/→, recall by name)	C	#13 → #18
Variable D (store with STO/→, recall by name)	D	#13 → #21
Variable E (store with STO/→, recall by name)	E	#13 → #22
Variable F (store with STO/→, recall by name)	F	#13 → #23
Variable G (store with STO/→, recall by name)	G	#13 → #24
Variable H (store with STO/→, recall by name)	H	#13 → #25
Variable I (store with STO/→, recall by name)	I	#13 → #26
Variable J (store with STO/→, recall by name)	J	#13 → #27
Variable K (store with STO/→, recall by name)	K	#13 → #28
Variable L (store with STO/→, recall by name)	L	#13 → #29

Variable M (store with STO/→, recall by name)	**M**	**#13 → #30**
Variable N (store with STO/→, recall by name)	**N**	**#13 → #31**
Variable O (store with STO/→, recall by name)	**O**	**#13 → #32**
Variable P (store with STO/→, recall by name)	**P**	**#13 → #33**
Variable Q (store with STO/→, recall by name)	**Q**	**#13 → #34**
Variable R (store with STO/→, recall by name)	**R**	**#13 → #35**
Variable S (store with STO/→, recall by name)	**S**	**#13 → #36**
Variable T (store with STO/→, recall by name)	**T**	**#13 → #37**
Variable U (store with STO/→, recall by name)	**U**	**#13 → #38**
Variable V (store with STO/→, recall by name)	**V**	**#13 → #39**
Variable W (store with STO/→, recall by name)	**W**	**#13 → #40**
Variable X (store with STO/→, recall by name)	**X**	**#13 → #41**
Variable Y (store with STO/→, recall by name)	**Y**	**#13 → #42**
Variable Z (store with STO/→, recall by name)	**Z**	**#13 → #43**
Variable θ (store with STO/→, recall by name)	**θ**	**#13 → #44**
VARS — variables menu	**VARS**	**#19**
WINDOW — set the graph view	**WINDOW**	**#2**

Y= — enter functions to graph	**Y=**	**#1**
ZOOM — zoom presets	**ZOOM**	**#3**
π (3.14159265…) as an exact constant	**π**	**#6 → #25**

Appendix C — Glossary

Ans
The result of the last calculation, reusable in the next.

AOS
Algebraic Operating System — operator precedence on a one-line display.

RPN
Reverse Polish Notation — operands first, then the operation; no = key.

Stack
The RPN registers X, Y, Z, T where numbers wait for operations.

TVM
Time Value of Money — n, i, PV, PMT, FV.

Natural display
Shows fractions, roots and powers as printed in textbooks.

Shift / 2nd
Prefix key that selects the function printed above a key.

Alpha
Prefix key that types the letter printed beside a key.

Radian
Angle unit; 2π rad = 360°.

Grad
Angle unit; 400 grad = 360°.

Engineering notation
Scientific notation with exponents that are multiples of 3.

Mantissa
The digits of a number in scientific notation.

Memory (M)

An independent register for running totals.

Mark-up

Increase over cost; margin is profit as a percent of selling price.

Grand total (GT)

Sum of all totals since the last clear.

nCr / nPr

Combinations / permutations of n items taken r at a time.

Factorial

n! = 1×2×…×n.

Reciprocal

1/x.

Standard deviation

Spread of data around the mean (σ population, s sample).

Key number

The small badge 1…N on each key in U-45 CALCULATORS, used throughout this manual.

Appendix D — Model notes & limits

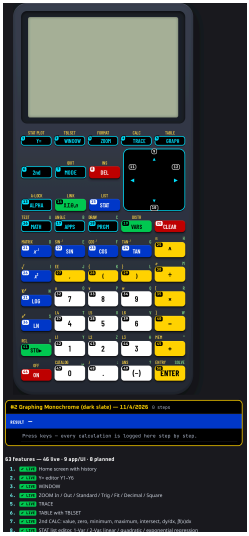
Monochrome graphing model. Home screen with history; Y= editor (Y1–Y6), WINDOW, ZOOM (box-free: In/Out/Standard/Trig/Fit), TRACE, GRAPH, TABLE/TBLSET, 2nd CALC (value, zero, minimum, maximum, intersect, dy/dx, ∫f(x)dx), STAT (list editor, 1-Var/2-Var and linear/quadratic/exponential regressions, scatter STAT PLOT), MATH (fractions, cube root, abs, round, nDeriv/fnInt), MATRIX, DISTR (normal/binomial/Poisson), ALPHA SOLVE (numeric solver). Programs, LINK and list literals {…} are not simulated.

Planned (not yet working)

- Programs (PRGM)
- LINK transfer
- List literals {…}
- Graph image export (PNG)
- Shareable function list link
- Dark screen theme
- Speech read-out of values
- Exam mode

U-45 CALCULATORS models are independent software recreations; they are not affiliated with any calculator manufacturer and carry no third-party trademarks.

Appendix E — Quick-reference card



#1 Y= ⇧ STAT PLOT	#2 WINDOW ⇧ TBLSET	#3 ZOOM ⇧ FORMAT
#4 TRACE ⇧ CALC	#5 GRAPH ⇧ TABLE	#6 2nd
#7 MODE ⇧ QUIT	#8 DEL ⇧ INS	#9 ▲
#10 ▼	#11 ◀	#12 ▶
#13 ALPHA ⇧ A-LOCK	#14 X,T,θ,n ⇧ LINK	#15 STAT ⇧ LIST
#16 MATH ⇧ TEST	#17 APPS ⇧ ANGLE	#18 PRGM ⇧ DRAW
#19 VARS ⇧ DISTR	#20 CLEAR	#21 x^{-1} ⇧ MATRIX
#22 SIN ⇧ SIN^{-1}	#23 COS ⇧ COS^{-1}	#24 TAN ⇧ TAN^{-1}
#25 ^ ⇧ π	#26 x^2 ⇧ √	#27 , ⇧ EE
#28 (⇧ {	#29) ⇧ }	#30 ÷ ⇧ e
#31 LOG ⇧ 10x	#32 7 ⇧ u	#33 8 ⇧ v
#34 9 ⇧ w	#35 × ⇧ [	#36 LN ⇧ ex
#37 4 ⇧ L4	#38 5 ⇧ L5	#39 6 ⇧ L6
#40 − ⇧]	#41 STO▶ ⇧ RCL	#42 1 ⇧ L1
#43 2 ⇧ L2	#44 3 ⇧ L3	#45 + ⇧ MEM
#46 ON ⇧ OFF	#47 0 ⇧ CATALOG	#48 . ⇧ i
#49 (−) ⇧ ANS	#50 ENTER ⇧ ENTRY	

U-45 CALCULATORS · Calculator #2 · cut out and keep